

Granular Instrumental Variables

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The Identification Problem in Macro

- **Challenge:** Most aggregate variables are endogenous (e.g., Prices vs. Quantities).
- **Typical Fix:** Ingenious natural experiments (rare) or Narrative shocks (hard to find).
- **The GIV Insight:**
 - Economies are "Granular" (Gabaix, 2011).
 - A few large firms/countries drive aggregate volatility.
 - Idiosyncratic shocks to *individual* large players are valid instruments for *aggregate* variables.

Benchmark Model: Supply and Demand

Demand by country i :

$$y_{it} = \phi^d p_t + \lambda_i \eta_t + u_{it}$$

Aggregate Supply:

$$s_t = \phi^s p_t + \epsilon_t$$

- p_t : Price
- η_t : Common demand shock
- u_{it} : Idiosyncratic demand shock
- **Goal:** Estimate ϕ^s .

How to build a GIV?

To isolate u_{it} from common factors η_t , use the difference between **Size-Weighted** and **Equal-Weighted** averages.

The GIV Definition (Simple Case)

$$z_t := y_{St} - y_{Et} = \sum S_i y_{it} - \frac{1}{N} \sum y_{it}$$

The Key Identifying Assumption

$$\mathbb{E}[u_t(\eta_t, \epsilon_t)'] = 0$$

Why granularity matter

- If $N \rightarrow \infty$ and sizes are equal, $z_t \rightarrow 0$ (No power).
- If S_1 is large, z_t remains volatile.
- **Power** comes from the Herfindahl index of the shares.

1. **Panel Regression:** Estimate y_{it} with time fixed effects to remove common shocks and extract residuals.

$$y_{it} = b_t + m^y C_{it}^y + \tilde{y}_{it}$$

2. **Factor estimation:** Use residuals and control variables to get factors.

$$\check{x}_i := x_i - x_E$$

$$\tilde{y}_{it}^e = \check{x}_i \eta_t^x + \tilde{u}_{it}$$

3. **OLS estimation:** Construct GIV instrument $z_t = S' Q y_t = \sum_i S_i \tilde{u}_{it}$ and run 2SLS.
4. **IV Regression:** Use z_t as an instrument for p_t in the aggregate supply/demand equation.

Optimal Weights

We seek weights Γ for $z_t = \Gamma' y_t$ that maximize instrument strength.

The "Z-filter"

Optimal weights Γ^* are the projection of sizes S onto the space orthogonal to factor loadings λ :

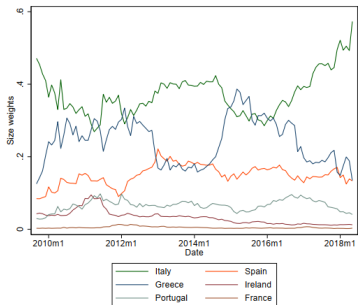
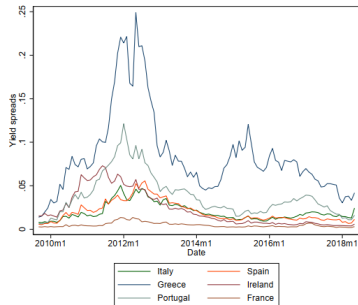
$$\Gamma^* = QS, \quad Q = I - \lambda(\lambda'\lambda)^{-1}\lambda'$$

- **Intuition:** Γ^* is the "closest" vector to S that completely removes aggregate shocks ($\Gamma'\lambda = 0$).
- Minimizes the asymptotic variance of the GIV estimator.

Application: Eurozone Yield Spillovers (following working paper)

- **Question:** If Italy's yield rises due to a local shock, how much does it spill over to France or Spain?
- **Size S_i :** Debt at risk ($\text{Debt} \times \text{Yield Spread}$).
- **Data:** 11 Eurozone countries (2009–2018).

Yield spreads and Size weights



- **Multiplier** $M \approx 1.5$: A 1 Euro idiosyncratic loss in one country leads to a 1.5 Euro loss in total Eurozone debt value.
- **Spillover** $\gamma \approx 0.33$: One third of the movement is pure contagion.
- **Narrative Check**: Top GIV shocks correspond to specific events (Greek bailout, Italian elections).

The GIV methodology has been rapidly adopted across several subfields:

- **Oil Markets: Baumeister & Hamilton (2023)**
- **Exchange Rates: Camanho, Hau & Rey (2022)**
- **Banking & Credit: Galaasen et al. (2021)**
- **Labor Markets: Schubert, Stansbury & Taska (2022)**
- **Financial Elasticity: Gabaix & Koijen (2022)**